

Shielded Outputs

Cryptography Benchmark Report

Hathor Network — Issue #1603 PoC

Pedersen commitments, Bulletproof range proofs,
and asset surjection proofs.

1. Overview

This report summarizes the performance of the shielded-outputs cryptography being prototyped for the Hathor Network (issue #1603). It quantifies the cost of constructing and verifying the cryptographic objects required by a fully shielded transaction: blinded Pedersen commitments, Bulletproof range proofs, and asset surjection proofs.

Three benchmark sweeps were run: (a) surjection proofs in isolation across an NxM grid of shielded inputs and outputs, (b) the full per-output workload (commitment + range proof + surjection) over the same grid, and (c) a mixed configuration with a fixed total of 64 inputs split between shielded and transparent.

Hardware: Intel Core i5-11300H (11th gen Tiger Lake, 4 cores / 8 threads, 3.1 GHz base / ~4.4 GHz boost), 11 GiB RAM, single-threaded execution under WSL2 (Linux 6.6) on Windows.

Cryptography is implemented in Rust and called from Python via PyO3 bindings (hathor-core / hathor-ct-crypto). Each grid cell is the average of 3 runs.

Memory consumption was not measured by the benchmark scripts (only wall-clock time), so the report does not report memory figures. Adding tracemalloc and getrusage hooks would be a small follow-up.

2. Methodology

Inputs and outputs are simulated using deterministic 32-byte token UIDs derived from an integer index. For each input, a raw asset tag is derived, a blinded generator $A = H_{\text{token}} + r_{\text{asset}} * G$ is constructed via `create_asset_commitment`, and a Pedersen commitment over that generator hides the amount.

Outputs reuse a token UID present in the input set so the surjection proof has a valid pre-image. Surjection construction is probabilistic (random subset sampling) and may fail; failed attempts are retried up to 5 times.

Times are measured with `time.perf_counter()` around the full Python loop. A separate "Rust-only" pass times just the FFI call (excluding Python loop overhead). The difference is small in this regime, so the report focuses on the full-loop numbers.

Grid points are taken from the powers of two intersected with $[1, \text{max}]$, plus the boundary value. For the surjection-only and full benchmarks the grid is 7x7 with N, M in $\{1, 2, 4, 8, 16, 32, 64\}$. The mixed benchmark sweeps the shielded input count s in $\{0, 1, 2, 4, 8, 16, 32, 64\}$ with the remainder $(64 - s)$ being transparent.

3. Surjection Proofs (Isolated)

This section measures the asset surjection proof in isolation: one proof per output, with the proof tying that output back to a domain of N input asset commitments. The Pedersen commitment and range proof are not included here.

3.1 Surjection: Creation Time

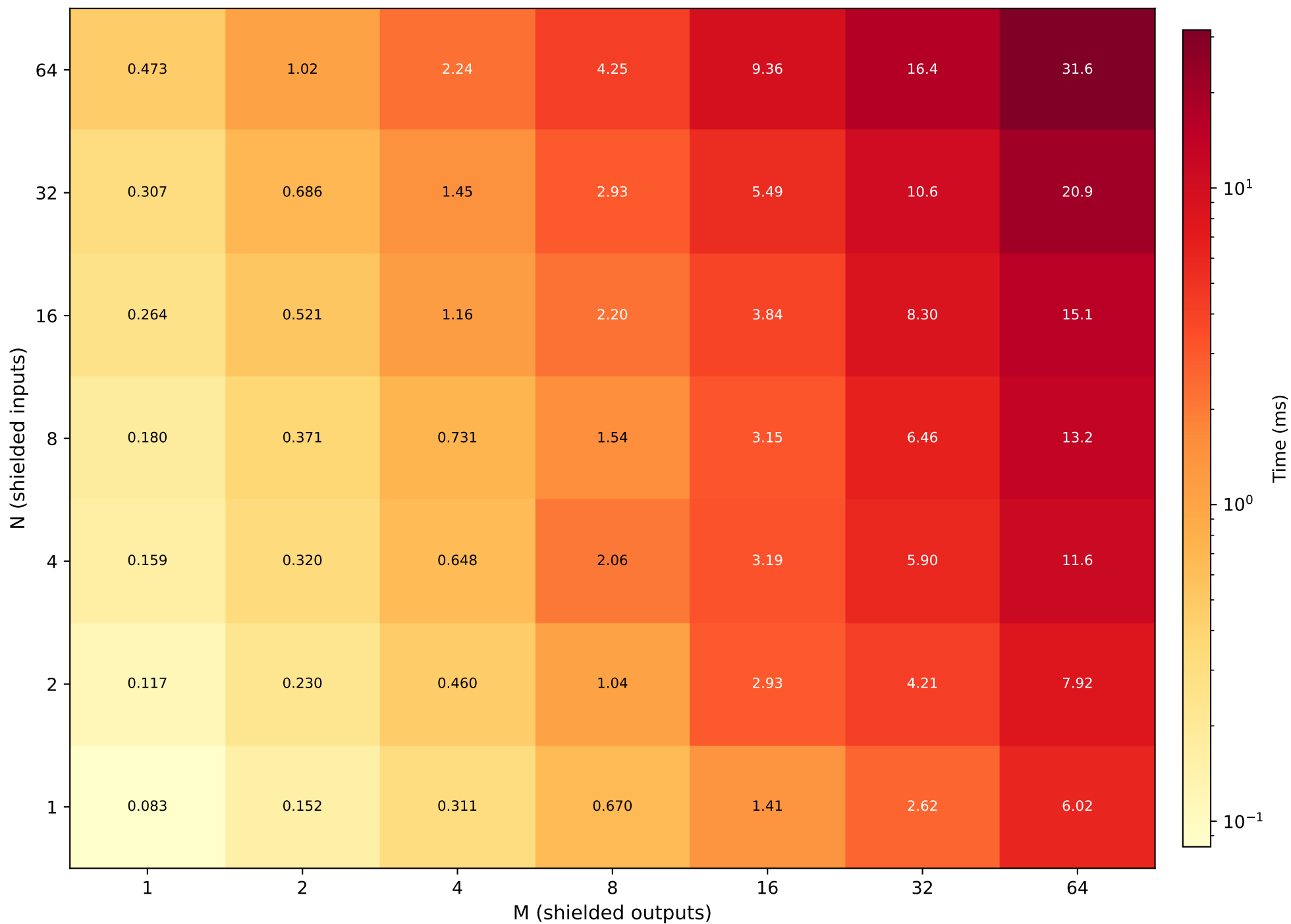
Range: 83.0 us (smallest cell) to 31.56 ms (largest cell), a 380x spread. Mean across the grid: 4.42 ms. At the smallest input set the per-output cost grows from 83.0 us ($M=1$) to 6.02 ms ($M=64$); at the largest input set, from 473.0 us to 31.56 ms.

Creation time scales close to linearly in M — each output gets its own proof, and per-output cost grows mildly with N because the proof has to position the codomain inside a larger domain.

The N dependence is sublinear: the Borromean ring signature inside the proof samples a small subset of the domain, so doubling N raises per-output cost by less than 2x. The largest cell ($N=64$, $M=64$) builds 64 proofs in roughly 32 ms, giving ~ 0.5 ms per proof at the densest point in the grid.

Surjection: Creation Time

Cell values in milliseconds. Color scale is logarithmic.



3.2 Surjection: Verification Time

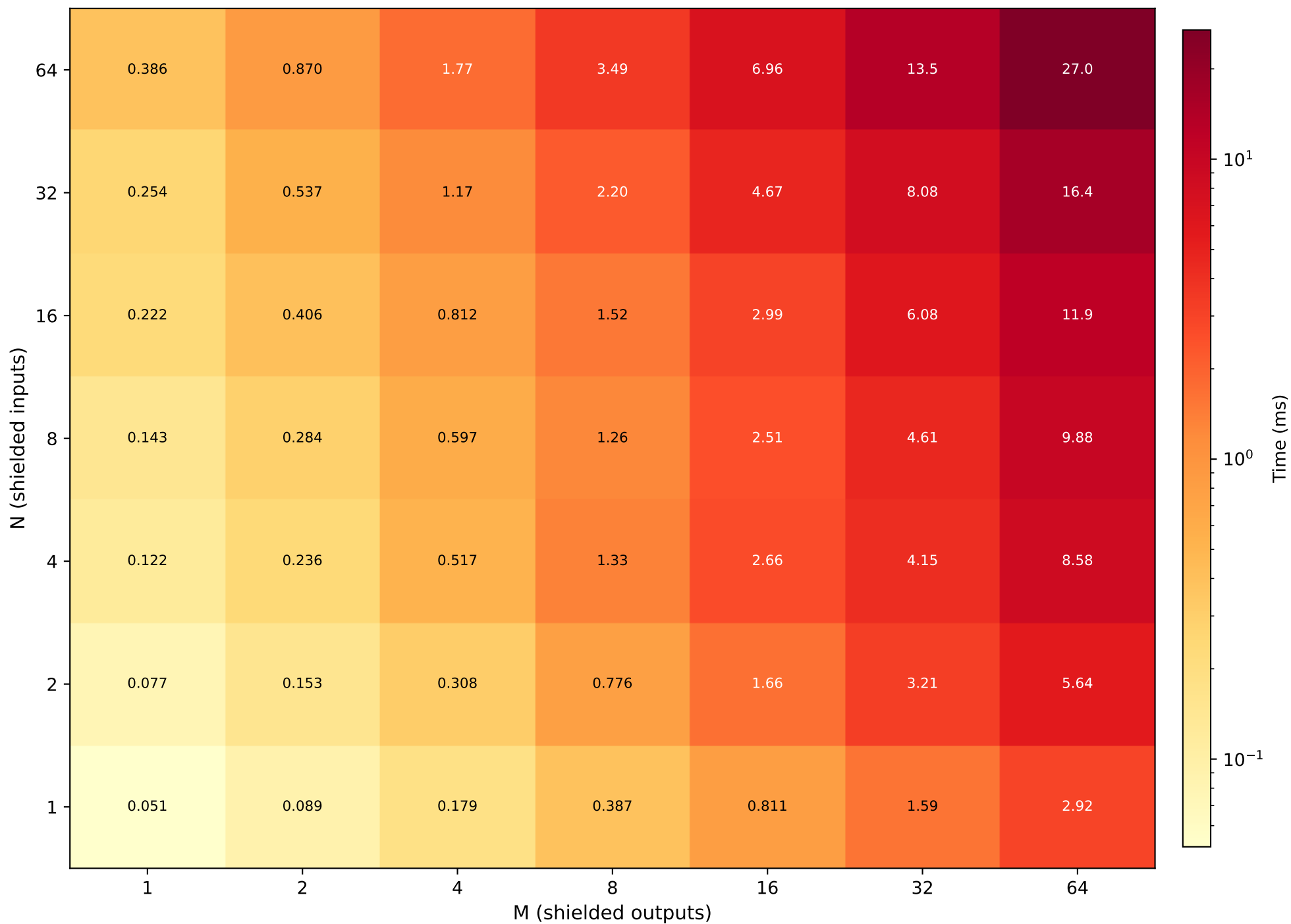
Range: 51.0 us (smallest cell) to 26.96 ms (largest cell), a 529x spread. Mean across the grid: 3.39 ms. At the smallest input set the per-output cost grows from 51.0 us ($M=1$) to 2.92 ms ($M=64$); at the largest input set, from 386.0 us to 26.96 ms.

Verification follows the same shape as creation but is consistently cheaper — typically 1.5x to 2x faster cell for cell. This is the expected asymmetry: the prover does the full Borromean ring construction while the verifier mostly performs scalar checks against the published proof.

For Hathor the verification side is what matters at network scale: every node verifies every transaction. At $N=M=64$ the full surjection batch verifies in ~ 27 ms.

Surjection: Verification Time

Cell values in milliseconds. Color scale is logarithmic.



4. Full Per-Output Workload

This section measures the three primitives that a real shielded output requires, in three pairs (creation, then verification): Pedersen + Bulletproof range proof, asset surjection, and the total. The range proof is fixed-cost per output (it does not depend on N), so it dominates at small input sets and gradually loses relative weight as N grows.

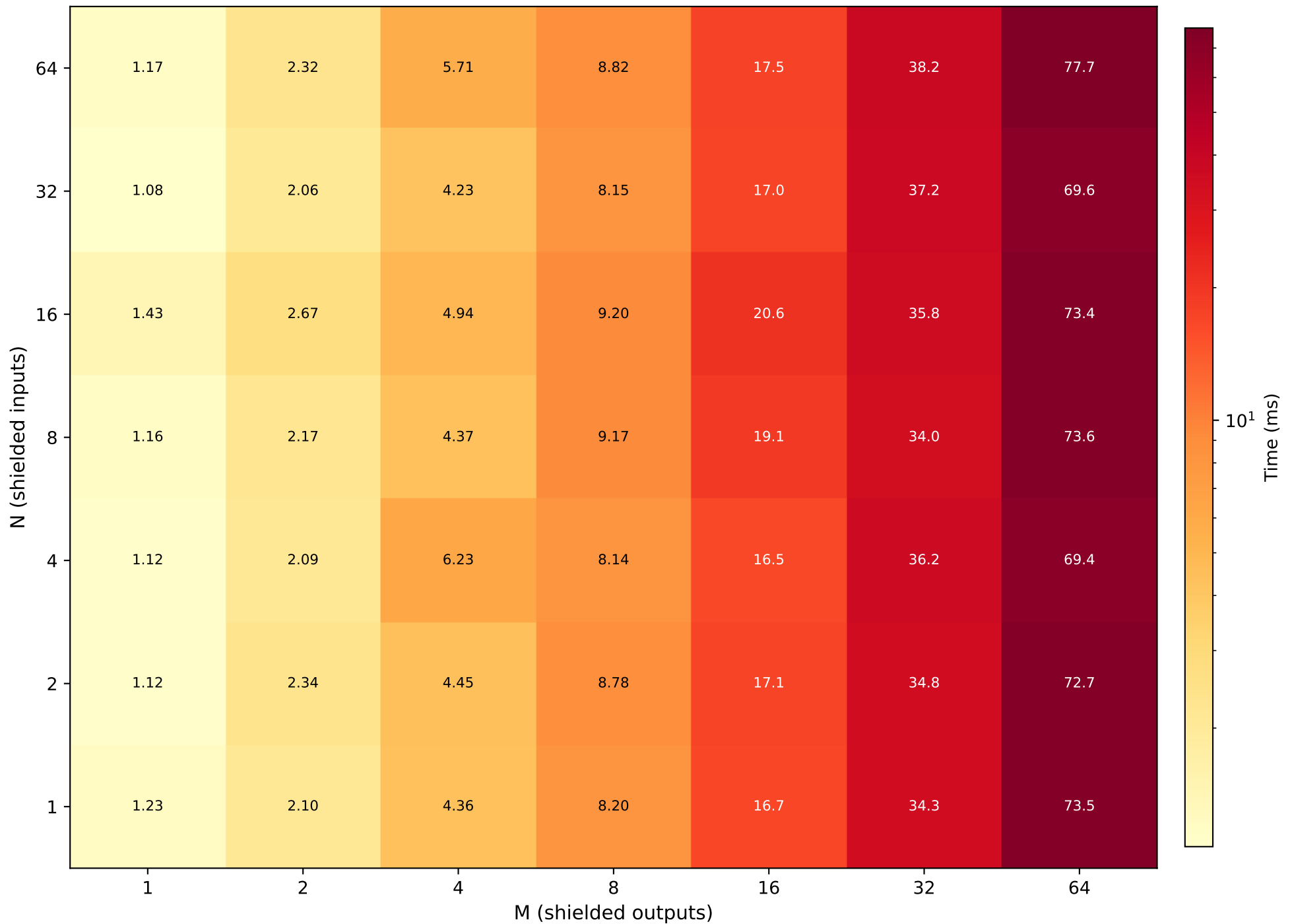
4.1 Pedersen + Range Proof: Creation Time

Range: 1.08 ms (smallest cell) to 77.72 ms (largest cell), a 72x spread. Mean across the grid: 20.48 ms. At the smallest input set the per-output cost grows from 1.23 ms (M=1) to 73.49 ms (M=64); at the largest input set, from 1.17 ms to 77.72 ms.

Creation cost is essentially flat in N (the Pedersen commitment and range proof do not consult the input set) and grows linearly in M. Each per-output Pedersen + Bulletproof costs roughly 1.2 ms to construct on this hardware, dominating per-output total cost up to $N \sim 16$.

Pedersen + Range Proof: Creation Time

Cell values in milliseconds. Color scale is logarithmic.



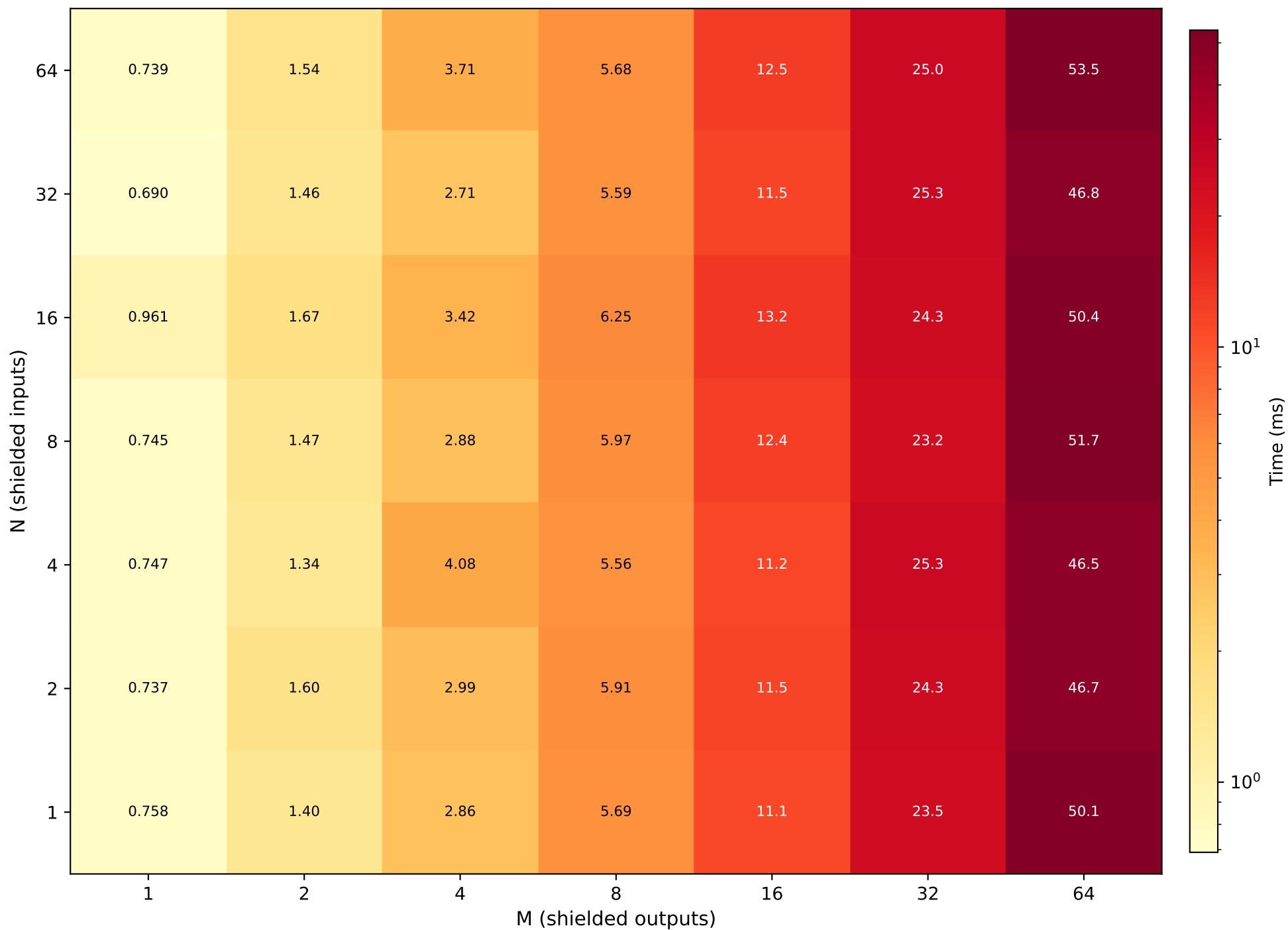
4.2 Range Proof: Verification Time

Range: 690.0 us (smallest cell) to 53.47 ms (largest cell), a 77x spread. Mean across the grid: 13.86 ms. At the smallest input set the per-output cost grows from 758.0 us ($M=1$) to 50.08 ms ($M=64$); at the largest input set, from 739.0 us to 53.47 ms.

Verification is also flat in N and linear in M . Per-output verify is around 0.8 ms — roughly 30% cheaper than creation, again the standard prover/verifier asymmetry for Bulletproofs.

Range Proof: Verification Time

Cell values in milliseconds. Color scale is logarithmic.



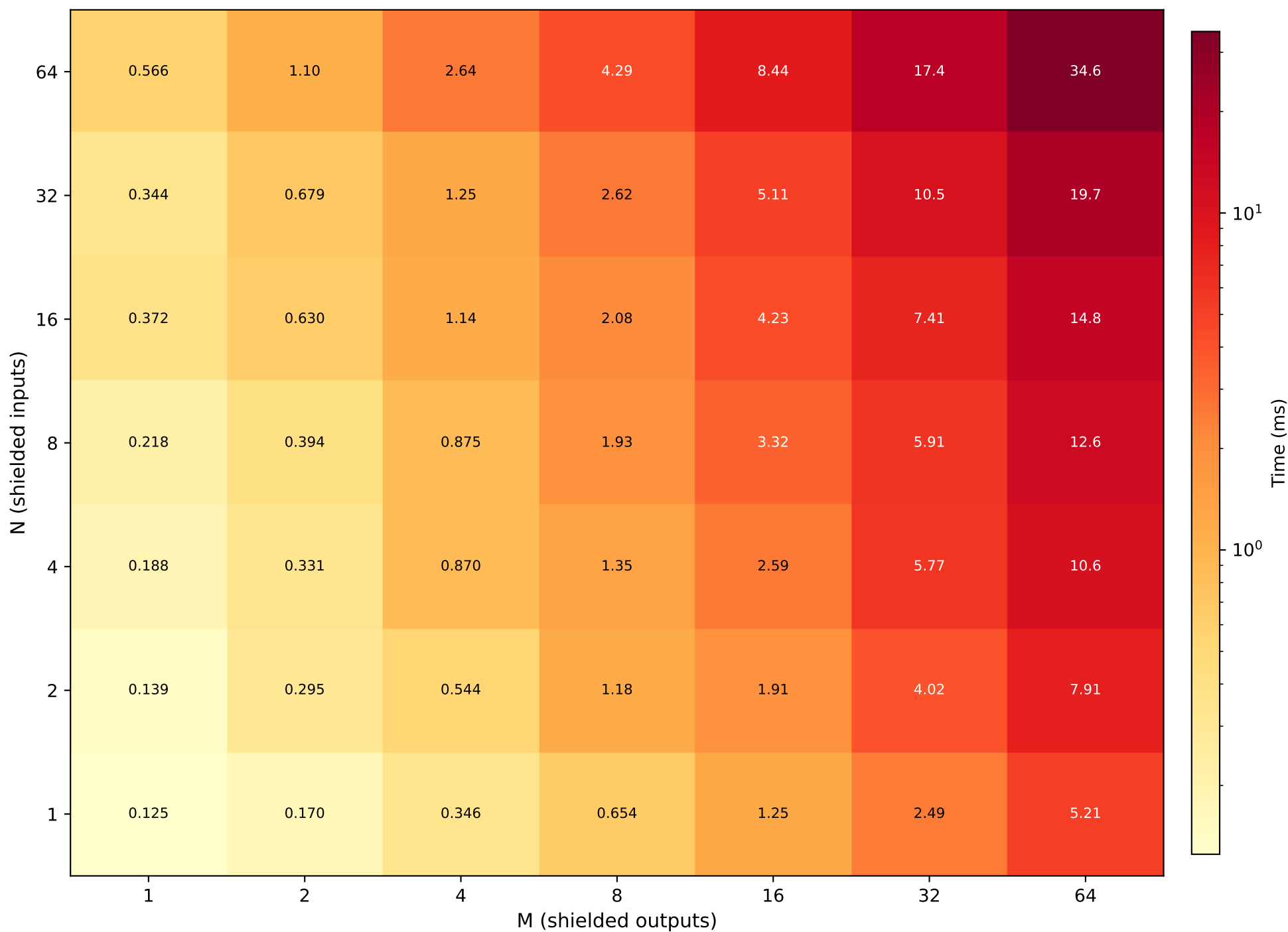
4.3 Surjection: Creation Time (Full Workload Run)

Range: 125.0 us (smallest cell) to 34.57 ms (largest cell), a 277x spread. Mean across the grid: 4.35 ms. At the smallest input set the per-output cost grows from 125.0 us (M=1) to 5.21 ms (M=64); at the largest input set, from 566.0 us to 34.57 ms.

Same primitive as section 3.1, re-measured during the full-workload run for consistency. Numbers track section 3.1 closely; minor variation comes from the probabilistic retry behavior and run-to-run jitter.

Surjection: Creation Time (Full Workload Run)

Cell values in milliseconds. Color scale is logarithmic.



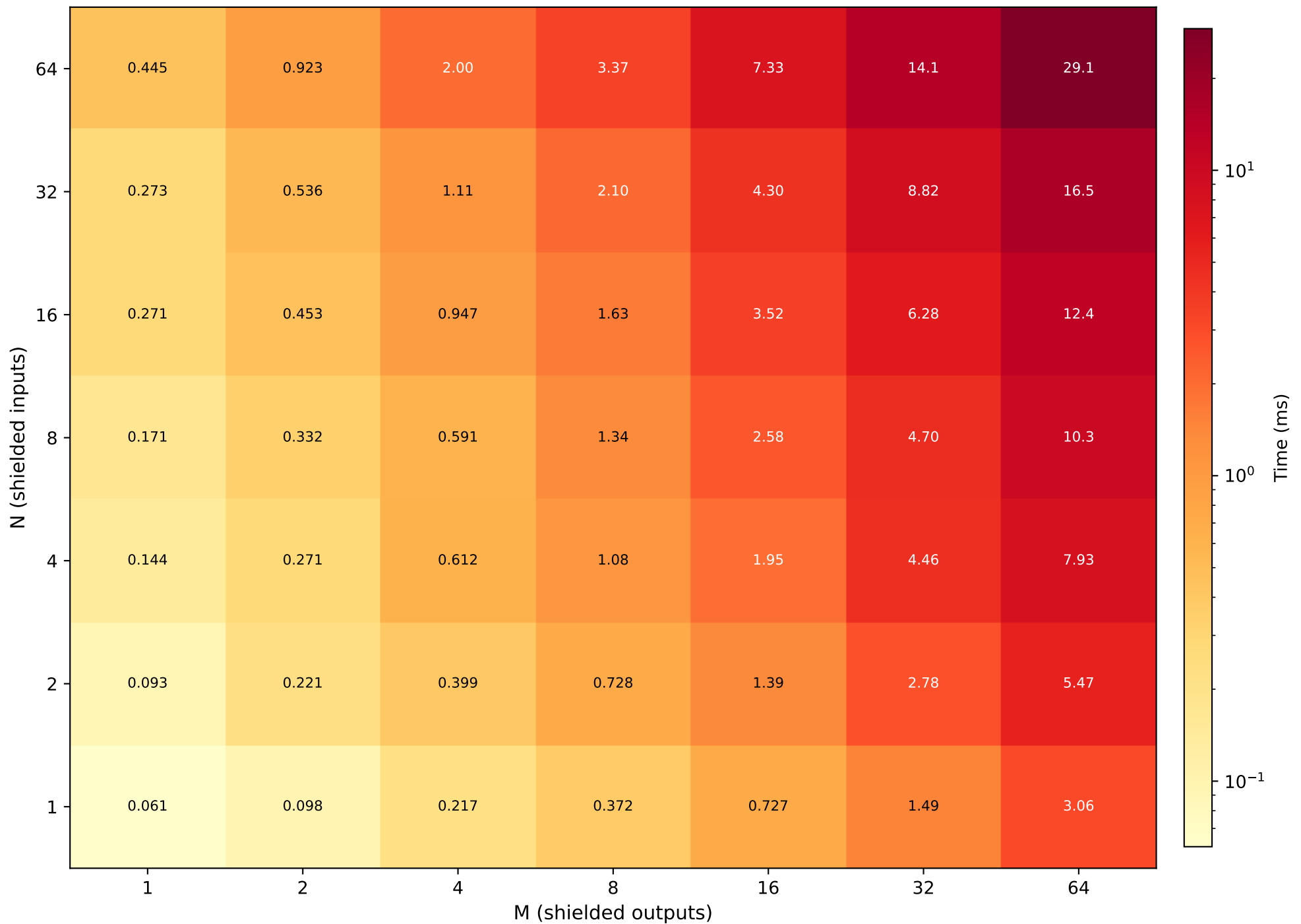
4.4 Surjection: Verification Time (Full Workload Run)

Range: 61.0 us (smallest cell) to 29.09 ms (largest cell), a 477x spread. Mean across the grid: 3.47 ms. At the smallest input set the per-output cost grows from 61.0 us ($M=1$) to 3.06 ms ($M=64$); at the largest input set, from 445.0 us to 29.09 ms.

Companion to 4.3; same observations as section 3.2. Verification stays cheaper than creation across the whole grid.

Surjection: Verification Time (Full Workload Run)

Cell values in milliseconds. Color scale is logarithmic.



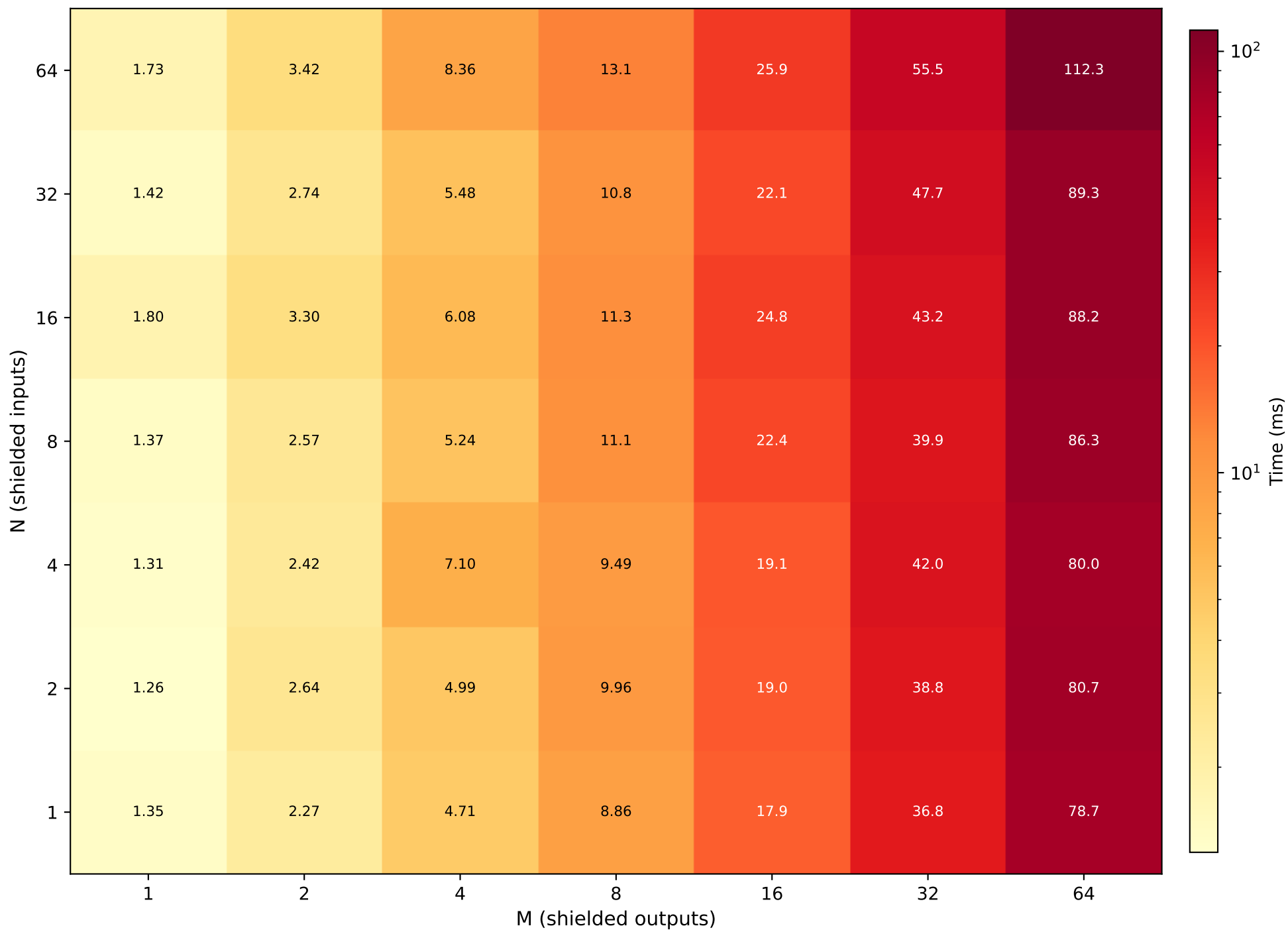
4.5 Total Per-Output Workload: Creation Time

Range: 1.26 ms (smallest cell) to 112.3 ms (largest cell), a 89x spread. Mean across the grid: 24.83 ms. At the smallest input set the per-output cost grows from 1.35 ms ($M=1$) to 78.69 ms ($M=64$); at the largest input set, from 1.73 ms to 112.3 ms.

Sum of Pedersen + range proof + surjection. At small N the range proof is the dominant cost; at $N=64$ the surjection overtakes it. The largest cell ($N=M=64$) constructs all three primitives for 64 outputs in roughly 110 ms total.

Total Per-Output Workload: Creation Time

Cell values in milliseconds. Color scale is logarithmic.



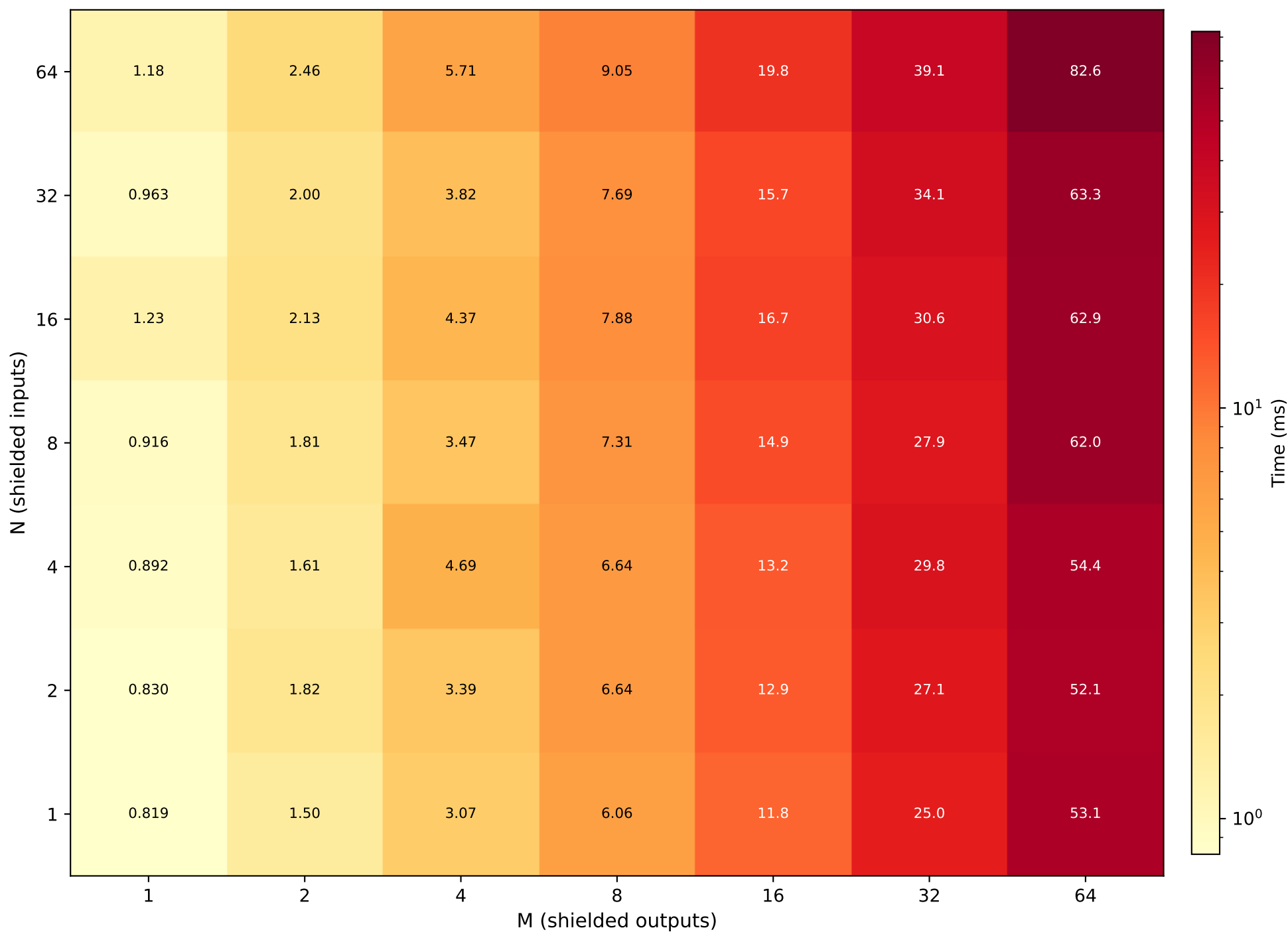
4.6 Total Per-Output Workload: Verification Time

Range: 819.0 us (smallest cell) to 82.55 ms (largest cell), a 101x spread. Mean across the grid: 17.33 ms. At the smallest input set the per-output cost grows from 819.0 us ($M=1$) to 53.14 ms ($M=64$); at the largest input set, from 1.18 ms to 82.55 ms.

Sum of all three verifications. Comfortably under 100 ms even at the top of the grid (~ 83 ms at $N=M=64$). This is the figure most relevant to network throughput, since every node will verify it.

Total Per-Output Workload: Verification Time

Cell values in milliseconds. Color scale is logarithmic.



5. Mixed Shielded / Transparent Inputs

A practical Hathor transaction is unlikely to be 100% shielded. This section fixes the total input count at 64 and sweeps the shielded share s in $\{0, 1, 2, 4, 8, 16, 32, 64\}$, with the remaining $64 - s$ inputs being transparent. Transparent inputs participate in the surjection domain via an unblinded generator (ZERO_TWEAK), so the total domain size stays at 64 regardless of the split.

5.1 Mixed: Creation Time

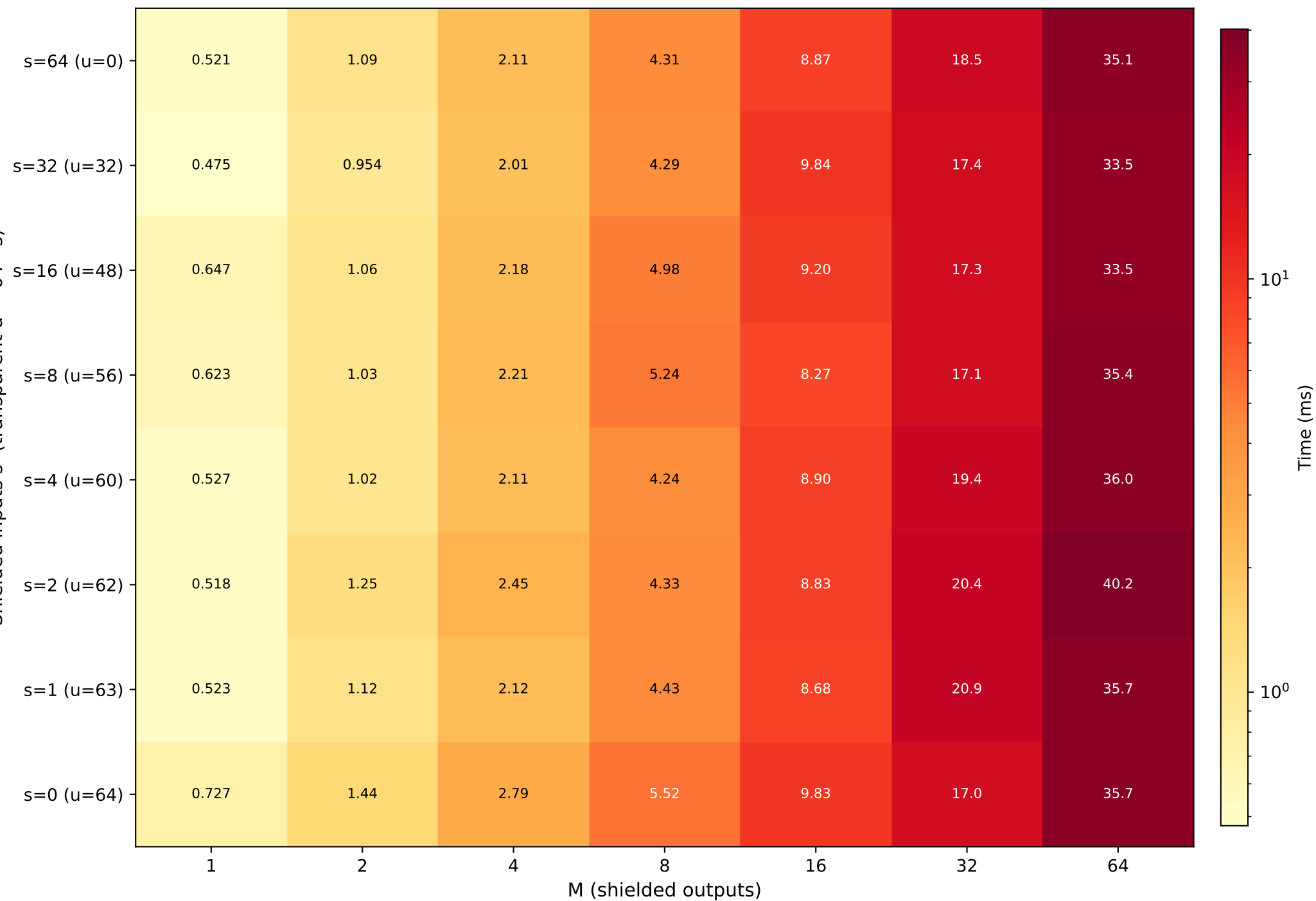
Range: 475.0 us (smallest cell) to 40.21 ms (largest cell), a 85x spread. Mean across the grid: 10.26 ms. At the smallest input set the per-output cost grows from 727.0 us ($M=1$) to 35.74 ms ($M=64$); at the largest input set, from 521.0 us to 35.14 ms.

Creation time is essentially flat across the shielded/transparent split for any given M . This confirms the design intent: surjection proof cost depends on the total domain size, not on how many domain members are blinded. A transaction author can mix shielded and transparent inputs freely without paying a per-share penalty.

The $s=0$ row (all transparent) is slightly slower for small M , likely because the codomain is still being constructed for shielded outputs while the domain is fully unblinded. The variation is within noise.

Mixed: Creation Time

Cell values in milliseconds. Color scale is logarithmic.



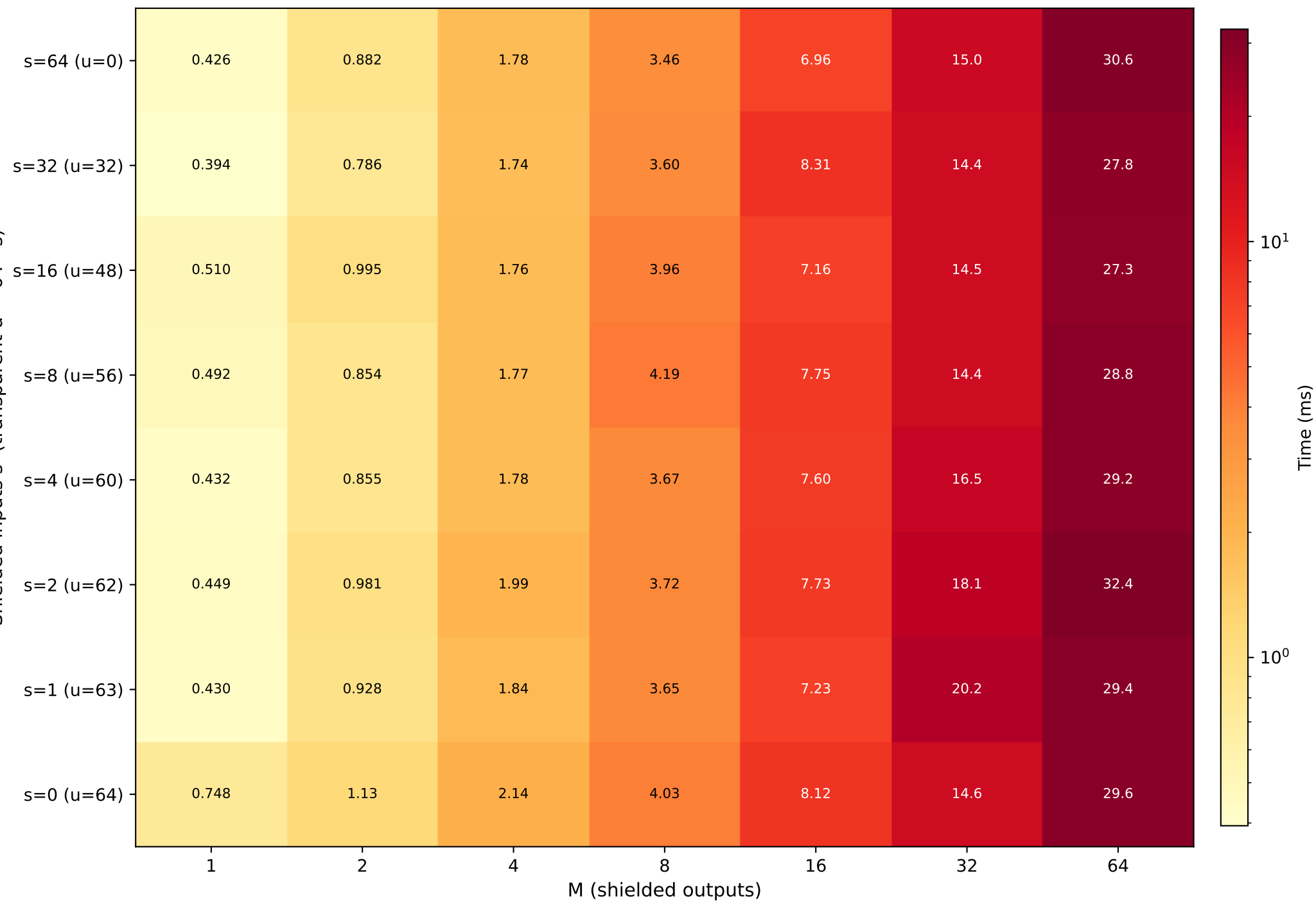
5.2 Mixed: Verification Time

Range: 394.0 us (smallest cell) to 32.40 ms (largest cell), a 82x spread. Mean across the grid: 8.58 ms. At the smallest input set the per-output cost grows from 748.0 us ($M=1$) to 29.64 ms ($M=64$); at the largest input set, from 426.0 us to 30.63 ms.

Same shape as 5.1: verification is flat in s and linear in M . Holding total input count fixed isolates the surjection cost from the shielded-share dimension, and the data confirms that the cost truly is independent of that share.

Mixed: Verification Time

Cell values in milliseconds. Color scale is logarithmic.



6. Conclusions

The shielded-outputs prototype is performant enough for the regimes tested. At the largest grid point (64 inputs, 64 outputs) the full creation completes in ~110 ms and full verification in ~83 ms on the test machine.

Verification is consistently 30-50% cheaper than creation, a healthy asymmetry for a network where each transaction is verified by many nodes.

Per-output cost is dominated by the Bulletproof range proof for small input sets and shifts toward the surjection proof as N grows past ~16. The ZERO_TWEAK trick for transparent inputs in the surjection domain has negligible cost overhead, making mixed shielded/transparent transactions cheap relative to fully-shielded transactions of the same total size.

Memory was not measured. Adding tracemalloc + getrusage instrumentation would close that gap; expected per-output footprint is small (a few KB of proof bytes plus transient scratch space inside the Rust library).